

**‘FIT NOURISH AI’ MOBILE APP FOR AI-POWERED
DYNAMIC MEAL GENERATOR & VISUAL
INGREDIENT IDENTIFIER**

Project ID –25-26J-386

Project Proposal Report

Ekanayake M.Y.R - IT22343352

BSc (Hons) in Information Technology Specialization in Information
Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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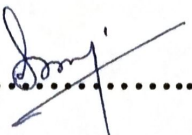
Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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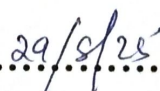
The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

.....


Signature of the supervisor

(Ms. Anjalie Gamage)

.....


Date

Abstract

The number of people suffering from health problems linked to their lifestyle, like obesity and diabetes, is growing worldwide. This shows a clear need to move away from general diet advice and towards a more personalized approach based on individual health data.

This project aims to fill a significant gap in research by creating a single, connected mobile app that combines two important features: recognizing food from a picture and creating a personalized meal plan. Currently, most tools either identify food or give diet plans, but they don't do both in a smart, integrated way.

Our main goal is to build a reliable app powered by Artificial Intelligence (AI). When you take a picture of your meal, the app will instantly give you a full report on its nutritional content and suggest healthy meals based on your personal needs.

To achieve this, our plan involves two key steps. First, we will train a special type of AI model that is great at understanding images. This model will accurately identify the food items in your picture and estimate their nutritional values, such as calories and fats. Second, we will use another AI model to take this nutritional information and match it to your unique health profile, which includes your goals and any dietary restrictions. This model will then provide useful, specific suggestions and healthy alternatives.

We will use existing public data for this project. Our work will follow a clear, step-by-step plan that covers everything from gathering data and building the AI models to putting the app together and testing it thoroughly. We expect to create an easy-to-use and highly accurate tool that helps people take control of their health in a new and effective way. This will provide a strong solution to the problems with traditional ways of tracking your diet.

Keywords: AI-powered, personalized meal plan, Machine learning, healthy alternatives, Image processing

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List of abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
BMI	Body Mass Index
CNN	Convolutional Neural Network
GPU	Graphics Processing Unit
ML	Machine Learning
SLIIT	Sri Lanka Institute of Information Technology
YOLO	You Only Look Once
LSTM	Long Short Term Memory
MAP	Mean Average Precision
MAPE	Mean Absolute Percentage Error

1.0 Introduction

1.1 Background

The global rise in lifestyle-related diseases, such as obesity, diabetes, and cardiovascular conditions, has underscored the need for more effective and personalized approaches to health and nutrition management. Traditional dietary planning often relies on a "one-size-fits-all" approach, providing generalized guidelines that fail to account for an individual's unique health conditions, genetic predispositions, and lifestyle choices [1]. This broad-stroke method frequently leads to suboptimal health outcomes and poor long-term adherence to dietary plans. The increasing demand for targeted and individualized nutritional solutions has paved the way for the integration of AI and machine learning (ML) in dietetics. AI-powered systems can analyze complex datasets to generate dynamic, data-driven nutritional recommendations, moving beyond static advice to provide real-time, adaptive guidance [1, 2].

A critical component of this new paradigm is the ability to automatically assess food intake. This has traditionally been a tedious process, relying on manual food diaries or self-reported data, which are often prone to inaccuracies. The emergence of computer vision has offered a transformative solution by enabling the automatic identification of food items and the estimation of their nutritional content from images. Early efforts in this domain faced significant challenges due to the immense visual diversity of food, including variations in presentation, portion size, and the presence of multiple food items in a single image [3]. However, recent advancements, particularly with Convolutional Neural Networks (CNN), have drastically improved accuracy. For instance, models like YOLOv8 have been successfully employed to detect and classify food items with high precision [4]. To provide accurate nutritional information, these systems must go beyond simple classification. A key innovation has been the development of methods to estimate portion size and volume from a two-dimensional image, allowing for more precise calculations of calories and macronutrients [4, 5]. This integration of image

processing with nutritional databases is the foundation of a modern food analysis system.

Building upon visual analysis, the next logical step is to map the extracted nutritional data to a user's specific health needs. This is where personalized dietary recommendation systems, driven by advanced ML algorithms, play a crucial role. These systems process a wide array of user-specific inputs, including biometric data like Body Mass Index (BMI) and glucose levels, along with health goals and dietary restrictions. Algorithms such as XGBoost and LightGBM have demonstrated significant success in classifying and recommending optimal meal plans, achieving high accuracy in tailoring suggestions to individual health profiles. A systematic review of AI applications in personalized nutrition highlights the effectiveness of such ML and deep learning models in improving clinical outcomes, from better glycemic control in diabetic patients to a reduction in gastrointestinal symptoms. Ultimately, the goal is to create a comprehensive, intelligent platform that not only identifies food and its nutritional value but also uses this information to provide proactive, personalized, and actionable dietary advice, thereby empowering users to take better control of their health [6].

1.2 Literature Survey

A growing body of research explores the application of AI in dietary management, specifically in developing systems that can analyze food from images and provide personalized nutritional guidance. This literature survey examines key research papers that form the foundation for an AI-powered dynamic meal generator and visual ingredient identifier. The following discussion delves into visual ingredient and nutritional analysis, personalized meal planning, and the overarching system design, drawing from seminal works in the field.

1.2.1 Visual ingredient and nutritional analysis from images

The initial challenge in developing such a system is the accurate identification of food items and the estimation of their nutritional content from an image. CNNs have emerged as the leading method for this task due to their ability to learn complex features from visual data. [3] introduced a hybrid CNN - Long Short Term Memory (LSTM) architecture. The CNN component extracts visual features from the food image, while the LSTM part processes these features sequentially to predict not only the food item but also a corresponding recipe. This approach underscores the potential of combining different deep learning models to perform multi-modal tasks, such as ingredient recognition and text generation. The authors utilized the publicly available Food-101 dataset and achieved a classification accuracy of 89.2% on food recognition, demonstrating the feasibility of using established datasets for training robust models. [4] presented an AI-powered food tracking system that uses YOLOv8, a state-of-the-art object detection model, to identify food items in images. Their system uniquely integrates a volumetric computation function to estimate portion sizes, a significant improvement over previous models that relied on 2D image analysis alone. By correlating the estimated volume with nutritional data from a comprehensive database, their model can accurately calculate calories and macronutrients. The researchers trained their model on a refined dataset of 95,000 instances across 214 food categories, achieving a mean average precision (MAP) of approximately 80% and highlighting the importance of combining advanced object

detection with methods for volumetric estimation to achieve precise nutritional analysis.

Another notable approach is detailed in [5], which demonstrates how multispectral imaging can enhance the accuracy of food classification and caloric estimation. This method fuses standard RGB images with images captured at various wavelengths in the ultra-violet and near-infrared bands. The researchers trained a CNN on this multispectral data, achieving a classification accuracy of up to 99.81% and a significant reduction in the mean absolute percentage error (MAPE) for caloric estimation. This work shows that incorporating richer, non-visual data can greatly improve the performance of CNN-based models for food analysis, though it requires specialized hardware.

Recent studies have also focused on addressing the challenge of limited data. For instance, [7] explores the use of transfer learning to adapt pre-trained models, such as ResNet and InceptionV3, for food image recognition. By using these models as a starting point and fine-tuning them on smaller, food-specific datasets, the researchers were able to achieve high accuracy without the need for massive computational resources. This approach is particularly valuable for developing applications that cater to culturally diverse cuisines where large, annotated datasets may not be readily available. The use of attention mechanisms has also been a focus, [8] demonstrates how these mechanisms can help models focus on the most relevant parts of a food image to better identify ingredients, even when the food is partially obscured or presented in complex arrangements.

1.2.2 Personalized meal planning with machine learning

Once the nutritional content of a meal is determined, the next step is to use this information to provide personalized recommendations. Machine learning (ML) models are essential for mapping a user's health profile to a suitable meal plan. A Personalized Meal Recommendation System was developed using a hybrid model that combines XGBoost and LightGBM classification and clustering algorithms in [2]. This system processes a wide range of user data, including BMI, age, glucose levels, dietary restrictions, and caloric requirements, to generate tailored meal plans. The authors' research hypothesizes that such an ML system can significantly enhance the accuracy and relevance of dietary suggestions, leading to better user adherence. The model achieved a prediction accuracy of 92%, demonstrating the effectiveness of using powerful boosting algorithms to handle complex, heterogeneous health data for personalized dietary recommendations.

Similarly, [1] provides a comprehensive look at the role of AI in personalized dietary recommendations. The review evaluates the effectiveness of various AI-generated dietary interventions, noting that most successful methods are based on ML, deep learning, and hybrid approaches. The paper presents compelling evidence that these interventions can lead to improved clinical outcomes, such as better glycemic control in diabetic patients and reduced symptom severity in those with irritable bowel syndrome. This review emphasizes that AI-driven systems are capable of surpassing traditional "one-size-fits-all" approaches by creating dynamic, data-driven recommendations that align with an individual's unique physiological and metabolic needs.

Further, [9] explores the use of collaborative filtering and reinforcement learning for personalized nutrition. Their system learns from user preferences and feedback to improve its recommendations over time, making it more adaptive and responsive to a user's changing tastes and health status. This approach is particularly useful for creating a dynamic and engaging user experience, as the system can continuously refine its suggestions based on implicit and explicit user data.

1.2.3 Comprehensive systems and user experience

Finally, the integration of these components into a cohesive application that provides a seamless user experience is crucial. Research in this area focuses on creating systems that not only perform the core tasks but also provide valuable feedback and recommendations. [6] describes a AI-driven multi-functional system that, while focused on recipe generation, lays a blueprint for a user-facing application. The system uses a CNN model for ingredient recognition and a Transformer-based model for recipe generation. This dual-model approach highlights how different AI architectures can be combined to perform multiple tasks, from image analysis to text generation. The system is designed to be a user-friendly web application, allowing users to upload an image and receive a recipe, which is a foundational concept for a broader meal recommendation system. [10] further explored user-centric design by developing an application that provides nutritional facts and alternative healthy meal suggestions. Their system used a ResNet-101 architecture for image classification and a decision-tree model for meal suggestions, reporting a strong correlation between the system's recommendations and user health goals. This work underscores the importance of a multi-modal approach and an intuitive user interface for creating a rich and interactive user experience. [11] emphasizes the need for user-friendly interfaces that simplify complex nutritional information and present it in an easily understandable format.

The literature collectively demonstrates that the project's goals are well-supported by established research. The use of advanced deep learning models like YOLOv8 and CNNs is a standard and effective method for visual food analysis, and ML algorithms such as XGBoost and LightGBM are powerful tools for personalizing meal plans based on health data. The key to a successful implementation lies in the careful collection and annotation of a suitable dataset, the integration of these models into a fluid and intuitive dashboard, and a robust validation process to ensure accuracy and reliability. A summary of key research papers, their architectures, and results is provided in Table 1.1.

Table 1.1: Summary of Key Literature in AI-Powered Dietary Management

Literature	Authors	Description	Architecture	Results
[4]	B. Panindre et al.	Developed a system to identify food, estimate portion size, and calculate calories from images.	Used YOLOv8 for object detection and a volumetric function.	Achieved MAP of approximately 80% for food item identification.
[5]	J. E. Oh et al.	Improved the accuracy of food and calorie estimation by using both regular (RGB) and special light (multispectral) images.	Used a CNN to analyze fused RGB and multispectral data.	Achieved a food classification accuracy of up to 99.81% and a reduced MAPE for calorie estimation.
[2]	P. C. Das et al.	Created a system that gives personalized meal plans based on a user's health profile and goals.	Used a hybrid of XGBoost and LightGBM algorithms.	Achieved a prediction accuracy of 92% for meal recommendations.
[1]	S. R. Lall et al.	A review of how AI can be used to give personalized diet advice and improve health results.	Reviewed various ML and deep learning methods.	The review noted that these systems led to improved health outcomes like better glycemic control and reduced symptoms of IBS.
[6]	R. D. D. S. R. Rajapaksha et al.	Created a system that takes a food image and generates a recipe.	Used a CNN for food recognition and a Transformer-based model for text generation.	This study served as a foundational blueprint for multi-modal systems, showing the feasibility of the architecture.

[3]	M.S. Lee et al.	Combined two types of AI models to recognize food and create recipes from images.	Used a hybrid CNN-LSTM architecture.	Achieved a food recognition accuracy of 89.2% on the Food-101 dataset.
[10]	Q. Wang et al.	Developed an app that gives nutritional facts and suggests healthy alternatives.	Used ResNet-101 for image classification and a decision-tree model for recommendations.	The system demonstrated a strong correlation between its recommendations and user health goals.
[7]	S. A. Khan et al.	Used existing AI models as a starting point to make new models that are good at recognizing food.	Used pre-trained models like ResNet and InceptionV3.	This research showed that high accuracy could be achieved with smaller, specialized datasets using transfer learning.

1.3 Research Gap

Despite significant advancements in AI-powered meal generators and visual ingredient identifiers, several key research gaps and challenges remain that require further investigation. Addressing these gaps is crucial for transitioning these technologies from promising academic prototypes to robust, reliable, and widely adopted consumer tools.

A primary research gap lies in the accuracy and generalizability of food image recognition models. While current models like YOLOv8 have demonstrated high accuracy in controlled environments, their performance often degrades significantly in real-world scenarios due to variations in lighting, food presentation, and partial occlusions [4]. A comprehensive review [4] and a study on multispectral imaging [5] highlight the difficulty of accurately estimating portion sizes and calories from single 2D images. Future research needs to focus on developing more sophisticated volumetric estimation techniques that can function reliably with standard mobile phone cameras, possibly by leveraging multi-view image processing or depth-sensing technologies. Furthermore, many existing datasets suffer from a lack of cultural diversity, being predominantly focused on Western cuisine, which limits the models' utility for a global user base and perpetuates algorithmic bias [6]. There is a critical need for larger, more diverse, and finely annotated datasets that include a wider range of regional dishes, ingredients, and preparation styles.

Another significant gap exists in the personalization aspect of these systems, specifically in the integration of complex, real-time health data beyond basic biometric information. While models that combine XGBoost and LightGBM show promise in mapping health profiles to meal plans, they often struggle with the dynamic and highly individualized nature of human physiology [2]. [1] points out the need for models that can effectively incorporate and interpret more granular data, such as real-time glucose monitoring, gut microbiome composition, and genetic information, to provide truly personalized and medically sound recommendations. The challenge here is not only in the data collection but also in the development of models that are both computationally efficient and capable of handling such high-

dimensional, heterogeneous data streams. Moreover, the "black box" nature of many deep learning models makes it difficult for both users and healthcare professionals to understand and trust the recommendations they provide [6], highlighting a need for more explainable and transparent AI frameworks in nutritional guidance.

Finally, the long-term efficacy and real-world impact of these AI systems are largely unexplored. While pilot studies may show promising short-term results in adherence and health outcomes, there is a lack of large-scale, long-term clinical trials that validate the effectiveness of AI-driven dietary interventions across diverse populations [1]. Research is needed to assess not only the technical performance of these systems but also their psychological and behavioral impact, including how they influence a user's relationship with food and their long-term health habits. Ethical considerations, such as data privacy and security, as well as the potential for algorithmic bias to exacerbate health disparities, remain critical areas that require more robust regulatory frameworks and research [1, 6]. Future work should focus on building comprehensive systems that not only address these technical challenges but also embed user-centric design and ethical principles at their core to ensure equitable and responsible deployment.

Table 1.2: Competitive Analysis

Paper	Data quality	Calorie, nutrient, & ingredient estimation	Warnings for unhealthy meals	Healthy food suggestions	transparency
[1]	NO	NO	YES	YES	YES
[2]	NO	NO	YES	YES	NO
[3]	NO	NO	NO	YES	NO
[4]	YES	YES	NO	NO	NO
[5]	YES	YES	NO	NO	NO
[6]	NO	NO	NO	YES	NO
[7]	YES	NO	NO	NO	NO

[8]	YES	NO	NO	NO	NO
[9]	NO	NO	YES	YES	NO
[10]	NO	YES	YES	YES	NO
[11]	NO	NO	YES	YES	YES
Fit Nourish AI	YES	YES	YES	YES	YES

1.4 Research Problem

The escalating global prevalence of lifestyle-related diseases, such as obesity, diabetes, and cardiovascular conditions, is a critical public health concern. A significant contributing factor to this crisis is the lack of effective, personalized dietary management. Traditional dietary tracking methods, which rely on manual food diaries or self-reported data, are often tedious, time-consuming, and prone to inaccuracies due to human error and recall bias. This creates a substantial gap between an individual's dietary habits and a precise understanding of their nutritional intake. Furthermore, the conventional "one-size-fits-all" dietary advice provided by generic health applications and public health campaigns fails to account for the unique physiological and metabolic needs of individuals, including their specific health profiles, dietary restrictions, and lifestyle goals. This lack of personalized nutritional guidance often leads to poor adherence and suboptimal health outcomes.

The core research problem this project addresses is the development of a comprehensive and integrated system that overcomes the limitations of current methods by seamlessly connecting visual food recognition with personalized health-based meal planning. Existing technologies often address these two aspects in isolation. On one hand, many AI food recognition systems can identify a meal and provide a list of ingredients and calories, but they lack the intelligence to interpret this data within the context of a user's health. On the other hand, personalized meal

planners may exist, but they typically require manual data entry, thereby retaining the inherent inaccuracies and burdens of traditional methods.

Therefore, the central research problem can be framed as follows: How can a single, cohesive system be designed and implemented that accurately identifies food items and their nutritional content from a user-uploaded image, and then dynamically maps this information to an individual's unique health profile to provide actionable, personalized, and healthy meal recommendations? The challenge lies in the effective integration of a high-performance CNN for image-based nutritional analysis with a robust ML model for personalized recommendation, all within a user-friendly and intuitive application interface. Successfully solving this problem will provide a novel and impactful tool for proactive health management, bridging the gap between passive food tracking and dynamic, personalized dietary intervention.

2.0 Objectives

2.1 Main objectives

The primary objective is to develop and implement a mobile-based, AI-powered system for personalized dietary management. This system will leverage image processing and ML to analyze food from images and generate health-conscious meal plans tailored to individual user profiles. The goal is to provide a comprehensive, user-friendly tool that offers instant, data-driven feedback and dynamic dietary recommendations to promote healthier lifestyles.

2.2 Specific objectives

To achieve the primary objective, the project will focus on the following specific objectives:

- **Data Collection and Management:** Collect and prepare annotated datasets of food images, nutritional information, and personal health profiles to train and validate the system's models.
- **Visual Analysis System:** Develop a CNN model to accurately identify food items, estimate portion sizes, and calculate nutritional values and calories from images captured by a smartphone camera.
- **Personalization Engine:** Create a machine learning model that maps a user's unique health profile, dietary restrictions, and goals to the nutritional data, providing personalized and actionable meal suggestions.
- **User Interface Design:** Design and implement an intuitive, user-friendly mobile application that allows for easy image uploads and provides clear, concise diagnostic results and recommendations with minimal technical expertise required.
- **System Validation and Optimization:** Conduct rigorous testing to validate the accuracy and reliability of the system, with a target accuracy rate of at least

90% for disease detection. Adjust model hyperparameters as needed to optimize performance.

- **Dynamic Recommendations:** Ensure the system can provide real-time recommendations, including suggestions for healthier alternative meals, to help users make informed decisions immediately after a food item is analyzed.

3.0 Methodology

3.1 Requirement gathering

The process of gathering requirements for this project will be a crucial first step. It will involve a detailed analysis of existing systems and a thorough review of past research to understand current capabilities, limitations, and user needs. The requirements will be collected through:

3.1.1 Past research analysis

An in-depth review of existing research papers and publications on food image recognition, personalized nutrition systems, and machine learning models will be conducted. This will help identify state-of-the-art methodologies and the technologies used in similar projects.

3.1.2 Analysis of existing systems

A competitive analysis of existing mobile applications for food recognition and diet planning will be performed. This will help in understanding the features, user interface designs, and a user's pain points, which will inform the design of our proposed solution.

3.1.3 Real-world scenarios

The project will also consider real-world user scenarios and behaviors related to dietary tracking to ensure the developed system is practical and addresses actual user needs.

3.2 Feasibility study

3.2.1 Technical feasibility

3.2.1.1 Knowledge on technologies

The development team has the necessary knowledge in key technologies, including:

1. **CNNs:** These are vital for food and ingredient recognition from images.
2. **ML models:** Techniques like XGBoost or LightGBM will be used to classify and recommend meals based on health profiles.
3. **Image Processing Libraries:** The use of libraries such as OpenCV is essential for preparing image data for the models
4. **Mobile Development Frameworks:** Cross-platform frameworks like Flutter or React Native will be used to build the mobile application

3.2.1.2 Knowledge on tools

The team is proficient in the necessary development and project management tools, including programming languages like Python, deep learning frameworks like TensorFlow, and IDEs like VS Code.

3.2.1.3 Data collection knowledge

The team possesses the skills to collect, annotate, and preprocess the required datasets for food images and user health profiles.

3.2.2 Schedule feasibility

The project will follow a detailed timeline over 12 months, which is considered feasible for a project of this scope. The project plan will be managed using a Gantt chart to ensure that all tasks are completed on schedule.

3.2.3 Economic feasibility

The system is a software-based solution, so the primary costs will be minimal. The main expenses will be for cloud computing services to train the models and host the application, which is considered an affordable and cost-effective solution.

3.3 System Analysis

3.3.1 Software Solution Approach

The development of this solution will follow a systematic approach:

1. **Data Collection:** Gathering and preparing extensive datasets of food images, nutritional information, and health profiles.
2. **Preprocessing:** Performing image preprocessing to prepare the food images for the CNN model.
3. **Feature Extraction:** The CNN will extract meaningful features from the food images, such as texture, shape, and color, which are critical for accurate food recognition.
4. **Model Training:** Training the CNN and ML models on the prepared datasets to identify food and generate personalized recommendations.
5. **Model Evaluation:** Evaluating the performance of the models using metrics like accuracy and precision to ensure they meet the project's requirements.

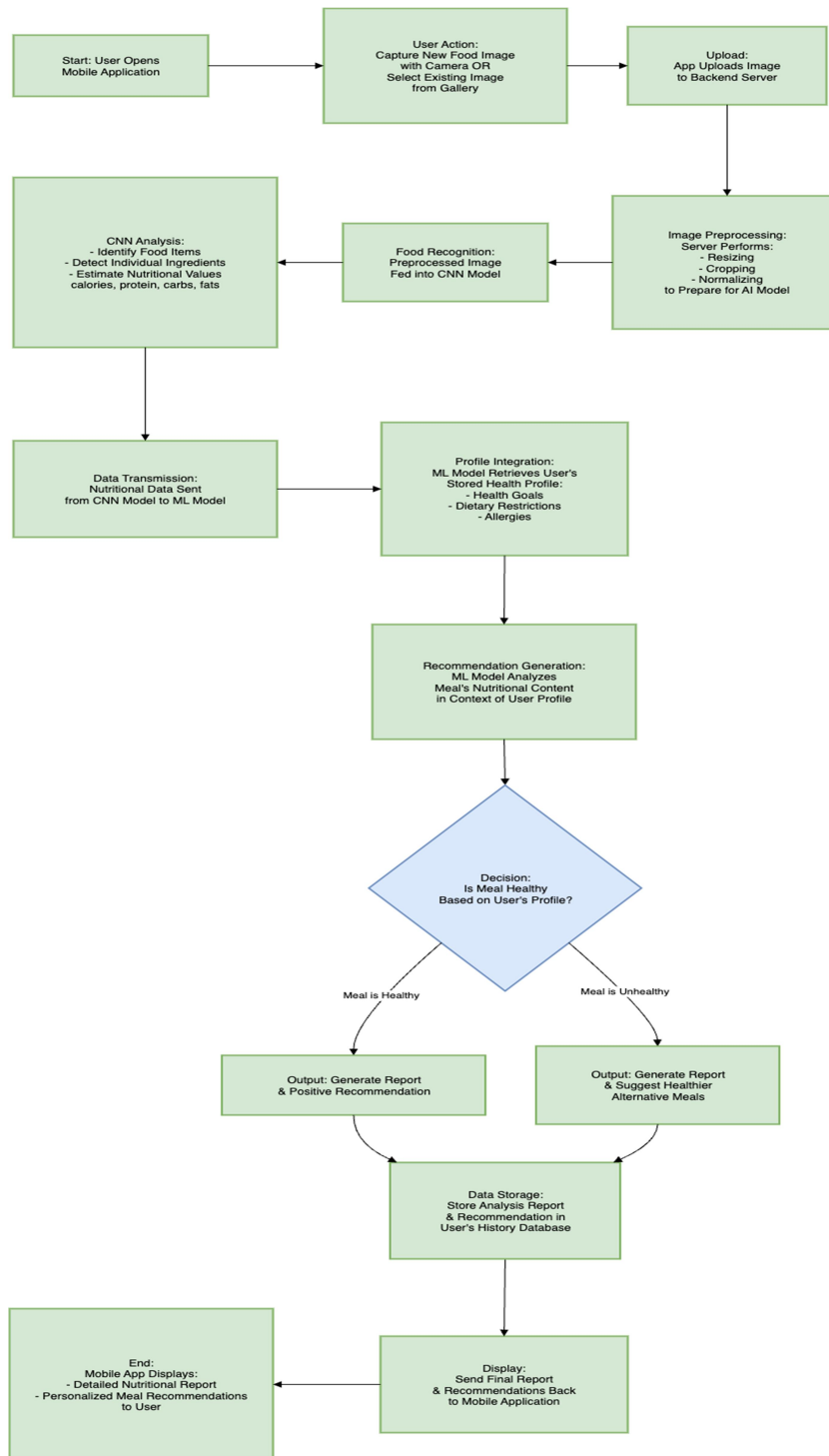


Figure 3.1: Flow Diagram of Food Recognition and Recommendation Engine

3.3.2 Tools & Technology

1. Programming Languages: Python
2. Frameworks: Flutter/React Native
3. Database Systems: A suitable database like PostgreSQL or MongoDB
4. Web Development: A backend framework for the API
5. UI Design: Figma for designing the user interface
6. IDE: VS Code
7. Version Control: GitHub
8. Collaboration Tools: Microsoft Teams

3.4 Project Requirements

3.4.1 Functional Requirements

1. The system shall allow users to upload images of meals from their mobile devices.
2. The system shall identify the food items and their ingredients in the uploaded image.
3. The system shall provide an estimated nutritional report for the identified meal.
4. The system shall use an ML model to assess the meal's healthiness based on a user's health profile and suggest healthier alternatives.
5. The system shall maintain a database to store user profiles and previous meal analyses.
6. The system shall support user registration and authentication.

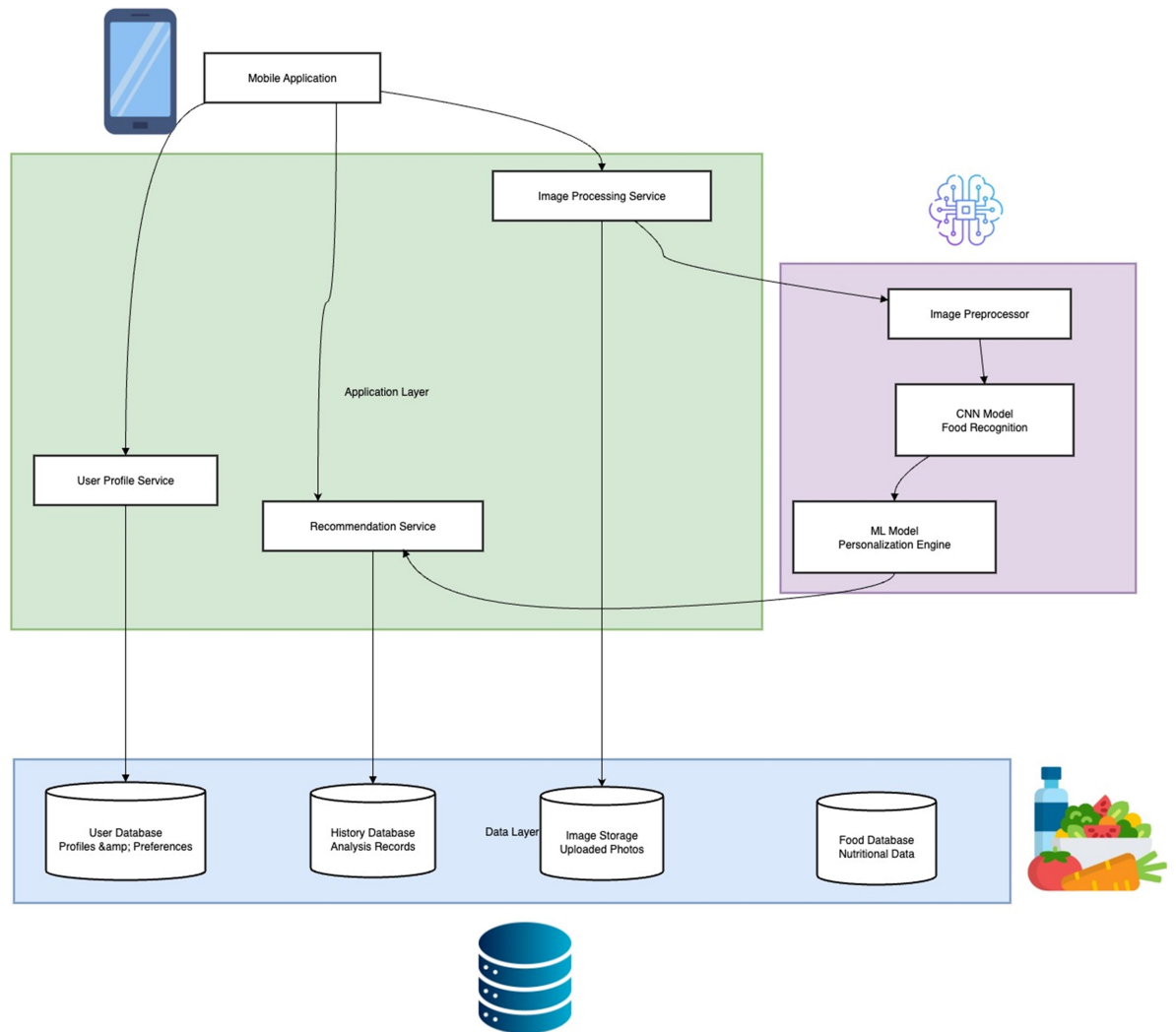


Figure 3.2: System Diagram

3.4.2 Non-Functional Requirements

1. The system shall achieve a food and ingredient identification accuracy rate of at least 90%.
2. The system's response time for providing a nutritional report shall not exceed 10 seconds.
3. The mobile application shall have an intuitive and user-friendly interface.
4. The system shall be scalable to handle a growing number of users and concurrent requests.

5. The system shall ensure data security and confidentiality of all user information.
6. The application shall be compatible with both Android and iOS platforms.

3.4.3 User Requirements

1. Users shall be able to easily capture or upload a food image using their mobile phone's camera.
2. Users shall receive a clear and easy-to-understand nutritional report for their meal.
3. Users shall be provided with actionable and relevant suggestions for healthier food replacements.
4. Users shall be able to create and update their personal health profile within the application.
5. Users shall have access to a history of their analyzed meals and recommendations.

3.4.4 System Requirements

3.4.4.1 Hardware Requirements

1. Smartphones: A mobile device with a camera resolution of at least 12 MP is required.
2. Server: A dedicated server with GPU support is necessary for training and running the deep learning models.
3. Internet: Stable and fast internet connectivity is required for data upload and retrieval.

3.4.4.2 Software Requirements

1. Backend: Python with TensorFlow/PyTorch and OpenCV.
2. Frontend: A cross-platform framework such as Flutter or React Native.
3. Database: A relational (e.g., PostgreSQL) or NoSQL (e.g., MongoDB) database.
4. Cloud Platform: Cloud services like AWS or Google Cloud for hosting and storage.

3.5 Use Cases (Tentative)

The following use cases, summarized in Table 3.1 and Table 3.2 outline the primary user interactions with the system and their corresponding functional goals.

Table 3.1: Use Case - Uploading a Meal Image

Use Case	Uploading a Meal Image
Actor	User
Description	The user takes a picture of a meal or selects one from their gallery to upload to the system for analysis.
Preconditions	The user is logged into the application.
Steps	1. Open the application. 2. Navigate to the "Upload Image" feature. 3. Capture a new image or select one from the gallery. 4. Submit the image for analysis.
Postconditions	The system provides a nutritional report and personalized recommendations.

Table 3.2: Use Case – Viewing Meal Recommendations

Use Case	Viewing Meal Recommendations
Actor	User
Description	The user reviews the healthy meal recommendations provided by the system
Preconditions	A meal image has been analyzed, and a nutritional report is available
Steps	1. The system displays the nutritional report after image analysis. 2. The user navigates to the "Recommendations" section. 3. The user views suggestions for healthy alternatives or meal plans

3.6 Test Cases (Tentative)

The following test cases, presented in Table 3.6, have been tentatively designed to validate the system's core functionality, performance, and accuracy

Table 3.3: Test Cases for the AI-Powered Dietary Management System

Test Case ID	Objective	Inputs	Expected Output
T.C.01	Validate the accuracy of food and ingredient identification	An image of a margherita pizza with basil, tomato, and mozzarella cheese	The system correctly identifies "margherita pizza," "basil," "tomato," and "mozzarella cheese" with an accuracy of at least 90%, along with a corresponding nutritional report.
T.C.02	Verify the system's ability to provide relevant healthy replacement suggestions	An image of a known unhealthy meal (e.g., a cheeseburger and fries) and a user profile with a "weight loss" goal	The system identifies the meal as unhealthy and suggests appropriate replacements (e.g., a grilled chicken sandwich with a side salad)
T.C.03	Ensure the mobile application's response time is within the non-functional requirements	A standard meal image uploaded from a mobile device under normal network conditions	The nutritional report and recommendations are displayed within 10 seconds of image submission

3.7 UI Designs

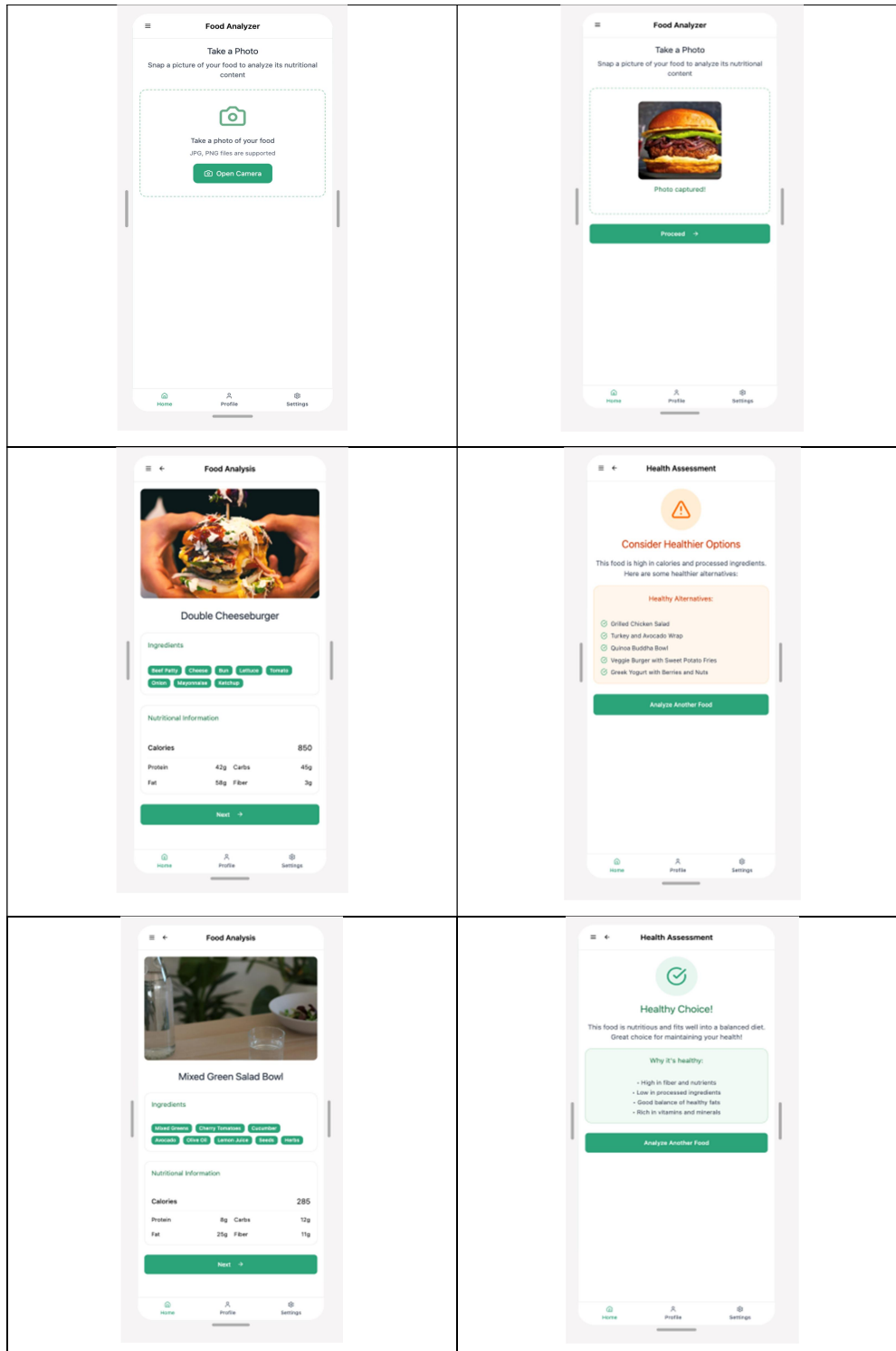


Figure 3.3: Figma UI designs

3.8 Timeline

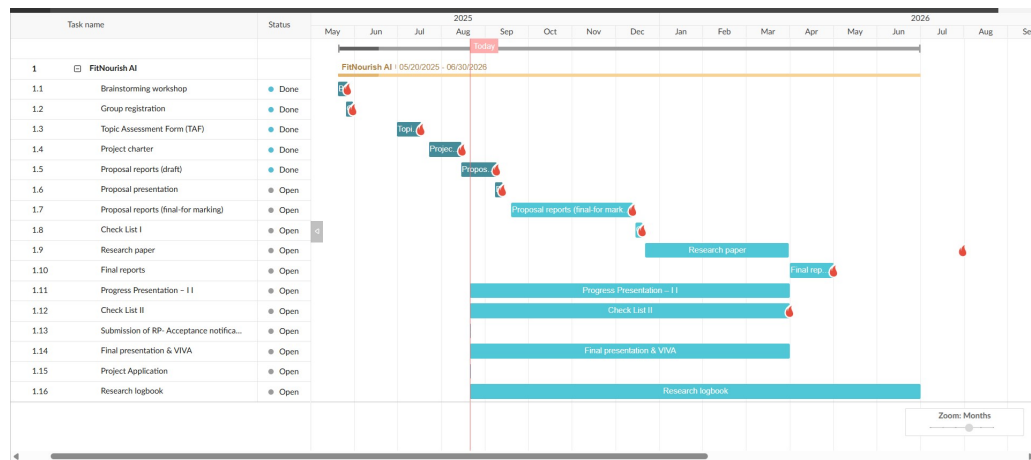


Figure 3.4: Gantt Chart

3.9 Communication Management Plan

The communications management plan ensures all stakeholders, including the project team and supervisors, have the information they need to perform their roles effectively. Clear and timely communication is essential for the project's success.

Communication Objectives

- Adequate: Provide the right information in the right format.
- Specific: Tailor information for the targeted audience.
- Sufficient: Give all the necessary information without overwhelming details.
- Concise: Keep communication brief and to the point.
- Timely: Address issues and updates at the appropriate time.

Communication Media

- Email: For formal communication, documentation, and key decisions.
- Phone Call/Chats: For quick, informal communication and urgent matters.
- Meetings: For detailed discussions, progress updates, and problem-solving.

4.0 Commercialization

The commercialization strategy for this project is centered on offering a highly specialized and impactful solution to a broad market. The target audience includes individuals focused on personal health and fitness, as well as businesses in the health, wellness, and food industries. The core value proposition of our system—providing real-time, personalized dietary feedback—is a unique feature that sets it apart from existing generic fitness and food-logging applications.

We envision a tiered commercialization model:

1. Direct-to-Consumer: The mobile application will be available on app stores (e.g., Apple App Store, Google Play Store) with a freemium model. A basic version with essential food recognition and nutritional analysis will be free, while a premium subscription will unlock advanced features. Premium features would include:
 - a. Personalization Engine: Access to the full ML model that provides tailored meal plans and healthy alternative suggestions based on a user's unique health profile.
 - b. Advanced Tracking: The ability to track meals over time and see trends in nutritional intake.
 - c. Expanded Database: Access to a more extensive database of food items, including diverse and regional cuisines.
2. Business-to-Business (B2B): The AI models and backend API can be licensed to other companies on a subscription basis. This would allow businesses, such as food delivery services, healthcare providers, or wellness platforms, to integrate our technology into their own applications. For example, a food delivery service could use our API to instantly provide nutritional information and healthy meal swaps for their menu items.

The project's commercial viability is strong due to its ability to address a critical market need for effective and personalized health solutions. The use of a simple,

mobile-based platform ensures a low barrier to entry for users, making the technology highly scalable and widely accessible.

5.0 Budget

A preliminary budget has been prepared to cover the essential resources required for the successful implementation and evaluation of the proposed system. Table 1 summarizes the anticipated costs.

Table 5.1: Budget Structure

Category	Description	Estimated Cost (LKR)
Hardware / Devices	Samsung Galaxy Watch 6 – for biometric data acquisition (heart rate, SpO ₂ , sleep, stress monitoring).	50,000
Software / Tools	Python, TensorFlow, PyTorch, Google Colab (open source / free tiers).	0
Cloud / Hosting	GPU-enabled cloud resources (Google Colab Pro / AWS EC2) for model training.	6,000
Datasets	Publicly available datasets (free) and premium access where necessary.	2,000
Documentation & Printing	Report preparation, printing, and binding.	3,000
Miscellaneous	Internet, electricity, and contingency costs.	4,000
Total Estimated Budget		65,000

5.1 Justification

The primary cost arises from the acquisition of the Samsung Galaxy Watch 6, which will be used to collect real-time biometric data including heart rate, SpO₂ levels, stress, and sleep quality. This device was selected due to its high accuracy of sensors, comprehensive health monitoring features, and SDK/API compatibility, which ensures seamless integration with the proposed nutrition and wellness system. Although lower cost alternatives such as fitness bands exist, their limited sensor reliability and restricted developer support make them less suitable for research validation.

Other costs are minimal since most of the required software tools are open-source.

However, limited expenditure is allocated for cloud-based GPU resources to accelerate deep learning model training, as well as for dataset acquisition, documentation, and contingency requirements. Overall, the total budget of LKR 65,000 is justified, with the smartwatch representing the critical hardware investment necessary to ensure data accuracy, system feasibility, and reliable evaluation outcomes.

6.0 Summary

The research project proposes to develop a comprehensive and innovative solution for personalized dietary management. The core problem addressed is the inefficiency and lack of personalization in traditional dietary tracking methods, which contributes to the global rise of lifestyle-related diseases. Our proposed solution is a mobile application that integrates image processing with machine learning to provide users with instant, data-driven nutritional analysis and tailored meal recommendations.

The methodology for this project is systematic and phased, beginning with the acquisition of a diverse and annotated dataset of food images and user health profiles. We will then develop and train two key AI models: a CNN model for accurate food recognition and nutritional estimation from images, and a ML model to serve as a personalization engine, mapping nutritional data to a user's specific health needs. These models will be integrated into a user-friendly mobile application prototype, which will undergo rigorous testing to ensure its accuracy and reliability.

The project is expected to deliver a highly accurate and functional tool that empowers individuals to proactively manage their health. By bridging the gap between passive food tracking and dynamic, personalized dietary intervention, this research has the potential to make a significant and positive impact on public health.

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